Alyza Abdullah

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EDUCATION

McGill University Montreal, Canada
Bachelor of Science, Kinesiology Aug 2023 – Apr 2027
Relevant coursework: Data Structures & Algorithms (Java; McGill School of Continuing Studies); Computer Programming for Life Sciences (Python); Principles of Statistics; Psychological Statistics; Research Methods; Biomechanics of Human Movement; Motor Development; Adapted Physical Activity; Pain; Neuroethics; Human Anatomy; Mammalian Physiology

RESEARCH EXPERIENCE

Human Pain Genetics Lab Research Assistant (PI: Dr. Luda Diatchenko) Montreal, Canada
Alan Edwards Centre for Research on Pain, McGill University Jan 2026 – Current
- Perform primary behavioral experiments and initial data analysis in translational studies that transfer plasma from long COVID and fibromyalgia patients into mice to model persistent pain.
- Run behavioral tracking and correct automated tracking output in analysis software to produce analysis-ready datasets.
- Co-author on two abstracts presented at the IASP 2026 World Congress on Pain; the long COVID abstract is also submitted to the QPRN Annual Meeting (Jan 2027).

MOGILab Research Assistant (PI: Dr. Jeffrey Mogil) Montreal, Canada
Alan Edwards Centre for Research on Pain, McGill University Aug 2025 – Dec 2025
- Began preclinical fibromyalgia pain research, conducting behavioral experiments and pain-related assays.
- Completed CMAR training; worked under approved animal research protocols and institutional compliance standards.
- Maintained experimental records and structured data organization for ongoing preclinical studies.

Language, Memory & Brain Lab Research Assistant Hamilton, Canada
Centre for Advanced Research in Experimental and Applied Linguistics (ARiEAL) May 2024 – Aug 2024
- Supported neuroscience research on language processing, working memory and auditory perception in adults.
- Designed and created auditory stimulus files for behavioral experiments, including tone sequences with controlled timing.
- Edited and standardized stimulus sets to ensure consistency across experimental conditions and participants.

Independent Scoping Review Montreal, Canada
Sole author May 2026 – Oct 2026
- Conducted a scoping review of motor profiles, measurement tools and interventions for movement difficulties in autism and developmental coordination disorder.
- Mapped how motor function is currently assessed, which interventions are used and where evidence of effectiveness is limited; findings directly informed the design of SYNC.

TECHNOLOGY AND DESIGN PROJECTS

SYNC · alyzaabdullah.github.io/sync Montreal, Canada
Designer and Developer Aug 2026 – Current
- Designed and built a play-based tablet tool that measures motor coordination in children, developed for autistic children with dyspraxia, with errorless, child-facing feedback and no scores shown to the child.
- Records target accuracy, hit rate, reaction and movement time, rhythmic timing, and raw orientation, accelerometer and gyroscope data with real timestamps for jerk-based smoothness analysis.
- Aim: a standardized measure for continuous tracking alongside intervention; next steps are test-retest reliability and validation against established motor assessments (BOT-2/BOT-3).

Circle Log · alyzaabdullah.github.io/circle-log Montreal, Canada
Designer and Developer, in partnership with Autisme sans limites Sep 2026 – Current
- Co-designing with program staff a practice-logging tool for a PEERS-based social-skills program for autistic young adults.
- Built participant goal-setting and practice logging, plus a facilitator dashboard showing group progress and opportunities to pair participants with shared goals.
- Staff pilot launching October 2026; participant rollout planned pending privacy and consent review under Quebec's Law 25.

PRESENTATIONS AND ABSTRACTS

- Surbey C, Wong M, **Abdullah A**, Zhou S, Falcone EL, Diatchenko L, Lima LV. *Plasma transfer from long COVID patients induces persistent mechanical hypersensitivity in mice*. Poster, IASP World Congress on Pain, Bangkok, Thailand, Oct 2026.
- Wong M, Lima LV, Vandermarcq T, Jahangiri Esfahani S, **Abdullah A**, Kawas L, Diatchenko L. *Modeling fibromyalgia in mice via patient plasma transfer: Persistent hyperalgesia and affective symptoms*. Poster, IASP World Congress on Pain, Bangkok, Thailand, Oct 2026.
- Surbey C, Wong M, **Abdullah A**, Zhou S, Falcone EL, Diatchenko L, Lima LV. *Plasma transfer from long COVID patients induces persistent mechanical hypersensitivity in mice*. Abstract submitted to QPRN Annual Meeting, Jan 2027.

MANUSCRIPTS

- **Abdullah A.** *Motor Coordination Difficulties in Autistic Children Aged 6–12: A Scoping Review of Praxis, Sensory-Motor Integration, and Interventions*. Preprint submitted; DOI forthcoming.

CLINICAL AND COMMUNITY EXPERIENCE

Adapted Physical Activity Practicum (EDKP 396) Montreal, Canada
McGill University Sep 2026 – Dec 2026

- Work one-on-one weekly with a young autistic child to design, implement and evaluate an individualized physical-activity intervention plan.

Neurological ICU Volunteer Montreal, Canada
Montreal Neurological Institute Oct 2024 – Current

- Shadow physicians and clinical teams during morning rounds in a neuro-intensive care unit, observing patient assessment.
- Provide sensory or cognitive stimulation to patients when recommended by physicians as part of patient care.
- Assist with routine clinical activities, including reading and tracking vital signs alongside nursing staff.

WORK, LEADERSHIP AND SERVICE

Research and Strategy Intern Remote
HerWILL Global · BLOOM: alyzaabdullah.github.io/bloom May 2025 – Current

- Designed and built BLOOM, an AI-literacy learning tool for the organization's programs.
- Support grant work, including research and writing for funding applications.
- Research global health, women's leadership and technology access; write briefs used for strategic planning and public-facing content.
- Help develop and coordinate mentorship programs, workshops and leadership initiatives.

Member and Mentor Montreal, Canada
Canadian Association of Global Health, McGill Chapter Oct 2024 – Current

- Mentor students on research involvement and academic planning; help plan and run workshops and events.
- Participate in case-based discussions on global health challenges, inequities and ethics.

SKILLS

Language Skills: Native in English and Bangla; basic Spanish and French
Technical Skills: Python; Java (object-oriented design, data structures and algorithms); AI-assisted software development (system design, testing, debugging); statistical analysis; Microsoft Excel
Certifications: CMAR research methodology and ethics workshops; Intro to Biosafety; WHMIS; HWM; OHRC; Emergency First Aid; Standard First Aid; CPR; National Lifeguard; Smart Serve